changes of I_{th} will arise only when the width of the MPI-channel is considerably enlarged, which requires a relatively high concentration of biomolecules. By contrast, with IR wavelengths, a much lower concentration may already affect I_{th}.

Local variations of the optical properties do not only affect the breakdown threshold but also lower the threshold sharpness (see section 2.3). In a similar way, it is expected that for UV-A and VIS ns pulses the transition between the small-bubble and luminescent-plasma regime is not as sharp in cells with inhomogeneous optical properties as in water. Experimental $R_{max}(E_L)$ data for different aqueous biological media showed that within a certain energy range, both small and much larger bubbles can be formed and the magnitude of the laser effect varies strongly from shot to shot [193,242]. Nevertheless, small bubbles could reproducibly be produced in murine intestine by slm 355-nm, 0.5-ns pulses with energies close to the bubble threshold [242]. Reliable low-density plasma formation by 0.5-ns, 532-nm slm pulses was also achieved in cell culture media [193].

Fluctuations of the breakdown threshold in biological media have also been reported for fs pulses, with dependencies on concentration of biomolecules, local optical properties, and staining [116,243,244]. However, sudden jumps to high-density plasma formation are here not observed because the $R_{max}(E_L)$ dependence is continuous [Fig. 5(c)]. Nevertheless, threshold fluctuations may result in fairly large variations of the effect size due to the steep slope of the $R_{max}(E_L)$ curves in the small-bubble regime.

5.2. Regimes and scaling laws of nonlinear energy deposition

The $R_{max}(E_L)$ curves of Fig 5(c) cover a large tuning range of nonlinear energy deposition in water. The volume of laser-produced bubbles in the (τ_L, λ_L, E_L) parameter space spans 11 orders of magnitude. With fs pulses and UV/VIS ns pulses, the bubble volume can be tuned by 8 orders of magnitude, when the pulse energy is varied by a factor of only 10-100. The specific shapes of the $R_{max}(E_L)$ curves in different regions of the (τ_L, λ_L, E_L) parameter space reflect the three breakdown scenarios introduced in section II. While pronounced differences between the scenarios are observed close to threshold, the energy dependence well above threshold is always the same. In high-density plasmas, most of the incoming laser energy is absorbed, and a constant fraction of the absorbed energy goes into bubble energy [Fig. 10(c)], which leads to $R_{max} \propto (E_L)^{1/3}$ [58,104]. In the following, we discuss the characteristic features of the different scenarios in the energy range between bubble threshold and high-density plasma regime.

Scenario 1 relates to fs breakdown at any wavelength. Here seed electrons from photoionization are

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readily available because of the high irradiance needed to complete the breakdown process. However, time constraints limit the possible number of doubling sequences during a femtosecond pulse and the strength of the ionization avalanche [37,113,115,146]. Therefore, the onset of breakdown is smooth and its endpoint can be precisely tuned by varying the laser irradiance.

Close to threshold, only a tiny fraction of the energy of the fs pulse is absorbed (Fig. 10). The R_{max} (E_L) scaling is here determined by nonlinear absorption through the interplay of MPI and AI, and by the growth of the plasma volume with increasing E_L. If nonlinear absorption was dominated by multiphoton effects, the slope of the $R_{max}(E_L)$ curves in the double-logarithmic graph should be similar to the cube root of the order of the multiphoton process needed for ionization, which is 3 at λ = 347 nm and 8 at λ = 1040 nm. Thus, close to threshold it should be m = $\sqrt[3]{3}$ = 1.44 at λ = 347 nm, and m = $\sqrt[3]{8}$ = 2 at λ = 1040 nm if the plasma volume was constant, and a bit steeper, when the growth of the plasma volume with E_L is considered. However, the actual slope is much steeper and amounts to m = 10.5 at λ= 347 nm, and m = 14.7 at λ = 1040 nm. The large slope steepness far beyond the value expected from MPI alone indicates a prominent contribution of avalanche ionization, in accordance with the results of our previous study on the wavelength dependence of fs breakdown [37]. Well above threshold, most of the incident laser energy is absorbed and the specific energy dependencies of MPI and AI cease to play a role. Here, the slope of the $R_{max}(E_L)$ curve gradually converges towards m = 1/3.

Scenario 2 describes the "big-bang" region at long pulse durations and wavelengths, where luminescent plasmas and large bubbles are produced already at threshold. Here, seed electron generation by photoionization is the critical hurdle for the occurrence of breakdown at IR wavelengths. Because of the high order of multiphoton processes, this requires a high irradiance. At the same time, avalanche ionization is powerful because its rate increases roughly proportional to λ^2 and many doubling sequences can occur during a ns pulse [113,115]. Due to the combination of high irradiance and AI rate, the onset of breakdown is abrupt and the ionization avalanche can progress very rapidly [22,231]. Already at threshold, about 50% of the incident energy is absorbed [223] and temperatures > 5000 K as well as pressures >7000 MPa are reached (Table 3), which are associated with bright plasma luminescence [Fig. 7(d)]. Correspondingly, the laser induced effects are highly disruptive [40,110,114,194,228]. For energies larger than $2\times E_{th}$, the scaling law can be approximated by $R_{max} \propto (E_L)^{1/3}$ [Fig. 5(c)], while close to threshold $R_{max} \propto (E_L - E_{th})^{1/3}$ provides better results [12].

Scenario 3 applies to slm UV and VIS ns pulses with smooth temporal shape. Here, multiphoton-

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generated seed electrons are readily available because of the large photon energies. Like in fs breakdown, this leads to a gradual onset of plasma formation, which makes it possible to create nanoeffects. The avalanche ionization rate is comparatively small at short wavelengths and the "slow" avalanche is, furthermore, inhibited by recombination. This results in a smaller conversion efficiency into bubble energy at threshold (Fig. 10) than for fs breakdown. Moreover, the slope of the $R_{max}(E_L)$ curves is smaller and the energy range in which nanoeffects and small microeffects can be produced, is broader. Close to threshold, the slope of the $R_{max}(E_L)$ curves first decreases because the growth of free electron density with increasing laser pulse energy is inhibited by recombination $\propto \rho_c^2$ [Eq. (3)]. However, this trend is counteracted by the exponential increase of ρ_{therm} with focal temperature [Eq. (4)]. When thermal ionization overcomes recombination, an abrupt transition to luminescent plasmas and large bubbles occurs [Figs. 5(c) and 7(b)]. In the high-density-plasma regime above the step in the $R_{max}(E_L)$ curve, the scaling law quickly converges to $R_{max} \propto (E_L)^{1/3}$, as in scenarios 1 and 2.

Let us now look at the transitions between scenarios 1 – 3. The "big bang" scenario 2, which characterizes breakdown at long wavelengths and pulse durations, merges into scenario 1, when the pulse duration is reduced below the thermalization time of CB electrons and thermal ionization ceases to play a role. At λ = 1064 nm, this occurs around τ_L = 30 ps [114,223]. With decreasing wavelength, seed electrons for AI become readily available, and scenario 2 changes into scenario 3. For τ_L = 2 ns, and $\lambda \geq 720$ nm, the onset of breakdown is still abrupt [36], whereas a smooth onset with nanobubble formation is observed for ns breakdown at λ = 532 nm. In the short wavelength range, scenario 1 with continuous tunability changes into scenario 3 exhibiting a step in the $R_{max}(E_L)$ curves, when the pulse duration becomes longer than recombination and thermalization times for CB electrons. Since the thermalization time is in the order of 20-30 ps [123,174], a step is expected for τ_L > 30 ps. The precise borders of the individual regions still need to be explored in future.

Figure 6 shows that slm UV ns pulses can produce nanoeffects at $NA \geq 0.3$, i.e. within a large range of focusing angles. The energy difference between bubble and BPL thresholds increases significantly, when the pulse duration is prolonged from ≈ 0.5 ns to several ns [Fig. 5(c)]. Because of the lower threshold irradiance at longer pulse durations (Table 2), the AI rate decreases and the inhibiting role of recombination becomes stronger. This leads to a broader tuning range of nano- and microeffects.

Historically, attention has been paid mostly to scenarios 1 and 2. Our discovery of scenario 3 for UV/VIS nanosecond breakdown became possible by a combination of several factors: Diffraction limited

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